



RDA

Development and Application of Insect Forecasting Models

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Introduction

Insect pests belonging to the families Pentatomidae, Alydidae, and Bruchidae pose significant economic threats to field crops and stored commodities in Korea. Their seasonal occurrence and population dynamics are strongly influenced by climatic conditions, particularly temperature. Climate change has increased concerns about shifts in insect development, emergence, and distribution, with Korea experiencing a notable rise in mean temperature over recent decades (Kwon et al., 2017). Under these changing climatic conditions, timely pest management is essential to minimize crop losses. Accurate forecasting models can provide early warnings of pest emergence and support effective decision-making for integrated pest management. Therefore, this study developed and evaluated linear and nonlinear weather-based forecasting models, including degree-day (DD) and phenology-based models, using thermal requirements to predict key seasonal events, particularly spring emergence, of economically important insect pests in stored-product and field-crop systems. The developed models were subsequently validated using seasonal insect-trap observations under field conditions.

Materials and Methods

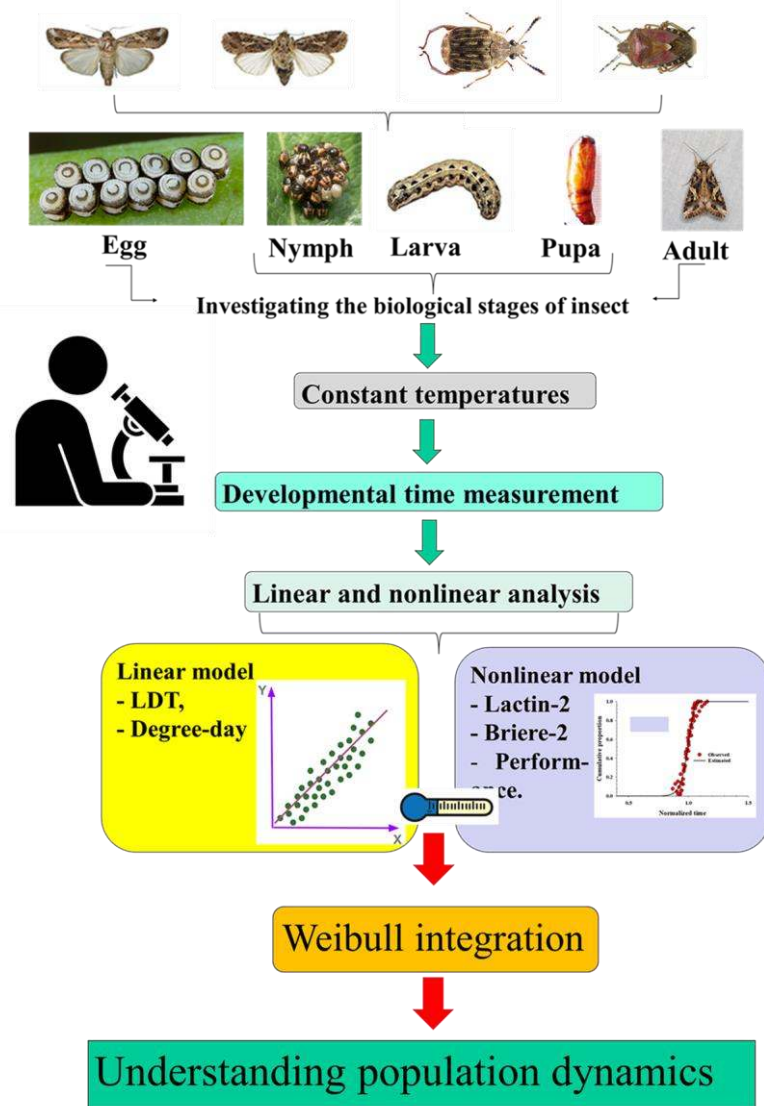


Fig 1. Overview of the experimental workflow for the temperature dependent development.

Results

Table 1. Total developmental time (day) of different insect species at different temperatures.

Insect species	Temperature (°C)									
	15	18	20	22	25	27	30	35	38	40
Spodoptera exigua	123.3		54	*	32.2	26.3	19.9	17.9		*
Spodoptera litura	106.3		42.4	*	30.4	30.1	26.0	22.5		Dead
Dolycoris baccarum			80.1	*	44.3	38.9	29.5	24.4		Dead
Callosobruchus chinensis	127.9	58.1		37.1	33.6	*	25.4	19.9	19.3	*

*: not included temperature

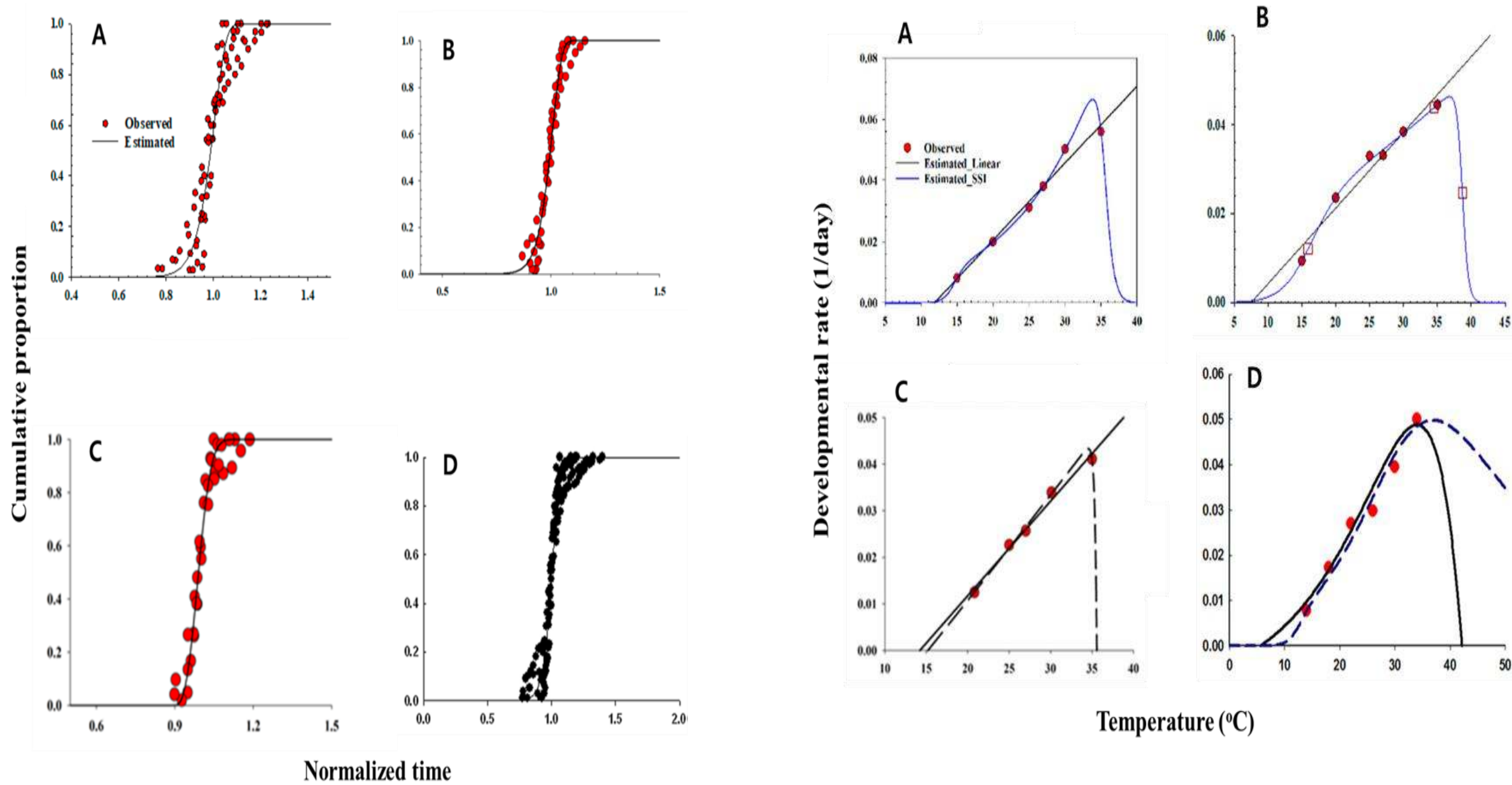


Fig. 2. Cumulative proportion and developmental rate of different insect species at different temperatures. (A) Spodoptera exigua, (B) Spodoptera litura, (C) Dolycoris baccarum, and (D) Callosobruchus chinensis.

Summary

We estimated thermal requirements of *S. exigua*, *S. litura*, *D. baccarum*, and *C. chinensis* using both linear and nonlinear models to create models that can predict its spring emergence and population trends. Our validated models show >95% accuracy of adult emergency in the fields. Utilizing our models, predicting the spring emergence of tested insect species in May-July.

Acknowledgment

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Table 2. Linear regression analysis of lower development threshold (LDT) and thermal constant (DD) of different insect species.

Insect species	LDT (°C)	K (DD)
<i>S. exigua</i>	11.88	397.27
<i>S. litura</i>	7.50	587.88
<i>D. baccarum</i>	14.22	492.22
<i>C. chinensis</i>	11.14	503.41

Table 3. Estimated annual voltinism of different insect species over different period in Miryang.

Insect species	Generation							
	2018	2019	2020	2021	2022	2023	2024	2025
<i>S. exigua</i>	5.24	5.1	4.97	5.35				
<i>S. litura</i>	5.3	5.23	5.16	5.47	5.42			
<i>D. baccarum</i>	3.27	3.16	3.08	3.36	3.3	3.54	3.77	
<i>C. chinensis</i>								

Table 4. Estimated annual date of spring emergence of different insect species over different period in Miryang based on a biofix of 1 January.

Insect species	Spring emergence date							
	2018	2019	2020	2021	2022	2023	2024	2025
<i>S. exigua</i>	05-Jun	10-Jun	06-Jun	09-Jun				
<i>S. litura</i>	28-May	31-May	26-May	26-May	24-May			
<i>D. baccarum</i>	01-Jul	05-Jul	27-Jun	04-Jul	28-Jun	01-Jul	29-Jun	
<i>C. chinensis</i>	Early June in the field and 43-44 days from egg hatching in the storage							

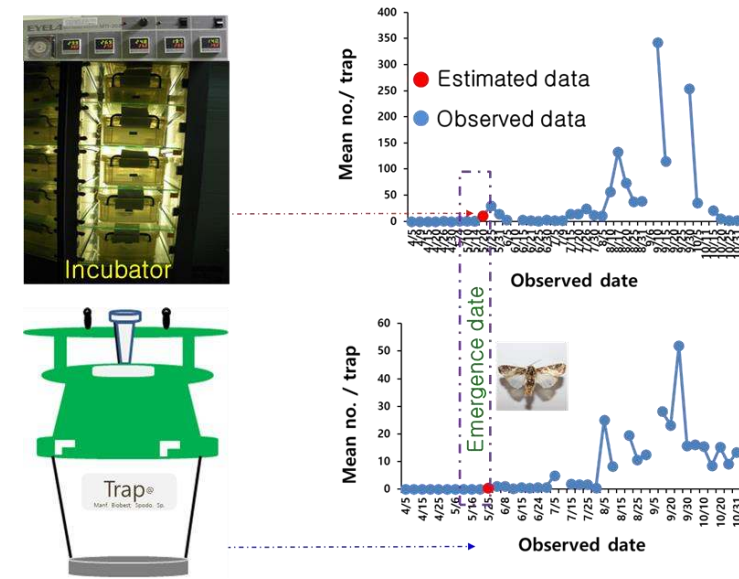


Fig 3. Validation of models from the field insect trapping data via different traps to forecast adult emergence in the fields.

Application & Managment

- As a result, this study provides important information on the temperature-dependent development of this polyphagous pest in Korea, which is expected to be useful for the prediction modeling of the distribution expansion and population regulation of tested insect species from a climate change perspective. The model developed in the present study could contribute to the development of integrated pest management strategies including spray timing, even with limited capacity of extrapolation from the laboratory-based parameter estimation (Maharjan et al., 2022, 2023, 2024, 2025).
- The findings of these studies can also be integrated with the AI technology for programming for developing system for forecasting the insect pests incidence in the crop fields.
- Models developed from this study can be applied to other wide rage of insect species to predict their spring emergence in the fields.

Reference

Kwon et al. 2017. [Journal of Asia Pacific Entomology](#). 20: 1249-1253.; Maharjan et al. 2017. [Journal of Stored Products Research](#). 72: 90-99.; Maharjan et al. 2022. [Insects](#). 13:747.; Maharjan et al. 2023. [Insects](#). 14: 412. Maharjan et al. 2024. [Bulletin of Entomological Research](#). 114: 606-612.; Maharjan et al. 2025. [Insects](#). 16: 1245.